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Networking and Information Technology Research and Development (NITRD) National
Coordination Office (NCO), National Science Foundation

Re: Development of an AI Action Plan. Request for input.

The Machine Intelligence Research Institute (MIRI) is a nonprofit based in Berkeley, California, founded in 2000. For over two decades, MIRI has worked to understand and prepare for the critical challenges that humanity will face as it transitions to a world with artificial superintelligence. Our technical and philosophical work helped found the field of AI alignment, and our researchers originated many of the theories and concepts central to today's discussions of AGI.

In our view, the field of AI is on a course to build AI systems that can substantially surpass humanity in all strategically relevant activities (often referred to as “artificial superintelligence”), with little to no understanding of how they function or ability to robustly steer and control their behavior. The development of such systems will ultimately lead to a loss of control, likely leading to human extinction. To avoid this risk, we believe it will be necessary to put in place coordinated global checks on AI development, capable of enabling, enforcing, and verifying restrictions on the development of superintelligent AI systems, until such a time as humanity's understanding of the relevant systems is sufficient to ensure they can be developed safely. As the global leader in AI, we believe the U.S. has the unique ability and responsibility to work with the international community to develop this capability, which we refer to as the *off-switch* (for a lack of a better succinct and nuanced term).

In this comment, we provide recommendations for the AI Action Plan covering expanding state capacity, maintaining America's AI leadership, coordinating with China to ensure global stability, and restricting the release of dangerous AI models. While these recommendations are motivated primarily by our focus on the risk of loss of control, we believe their value does not hinge on agreement with our views. These measures would strengthen U.S. AI governance capacity and provide crucial flexibility for policymakers across a spectrum of potential risk scenarios.

1. Our Perspective

The default consequence of artificial superintelligence is human extinction.

The timeline to ASI is highly uncertain, but probably not long. MIRI would not be surprised if ASI is developed during the present administration. It could arrive quickly following the development of AI systems which can automate AI research, which could plausibly happen by 2027.

AI labs are aggressively rolling out systems they don't understand. Deep learning techniques result in systems where, even though AI developers see what goes in and what comes out, they can't tell you why these systems make a given decision.

Developers appear far from able to give ASIs goals of the developer's choosing. The field seems very unlikely to find a robust solution to this problem in time to apply it to ASI.

An ASI with unintended goals would be very likely to cause our extinction. Unless it has goals of the developer's choosing, ASI will put our planet to uses incompatible with our continued survival. This extreme outcome doesn't require any malice, resentment, or misunderstanding, only a vastly superior intelligence in pursuit of its own goals.

These concerns are shared by [many experts who similarly believe](#) advanced AI could threaten human extinction.

This is not a case where we just need the “right” people to build it before the “wrong” people do. Therefore, [we believe](#) humanity, led by the United States, must be prepared to take a different approach: halting frontier AI development, since our survival likely depends on delaying the creation of ASI, as soon as we can for as long as necessary. Such a halt would be a coordinated moratorium on dangerous AI, potentially lasting decades. It would need to be collectively enforced globally by nations who share an understanding of the risks and a willingness to cooperate to protect against them.

1.1 The Off-Switch

At the present time, there is not sufficient consensus or political will to implement a halt to frontier AI development. Given this reality, our focus is on advocating that the world retain the ability to halt development should this change. We believe nations should begin building this capacity now—what we call an *off-switch*—before it's too late to implement effectively.

Building the off-switch entails establishing the technical, legal, and institutional infrastructure required to implement global checks on AI development, capable of enabling, enforcing, and verifying restrictions on the training, deployment, and proliferation of AI systems, up to and including the



ability to impose a halt on progress towards ASI.

A comprehensive off-switch would serve multiple functions: it would include emergency response protocols for acute risks (such as shutting down rogue autonomous systems), while also enabling longer-term policy and technical solutions to implement precisely targeted, durable, reversible restrictions on AI activities of concern as needed.

We want to emphasize that America's technological leadership and an off-switch are complementary goals. The off-switch represents strategic optionality—preserving the ability to halt dangerous AI development if and when risks become unacceptable, while maintaining innovation leadership. This capacity would also address a broader set of national security concerns beyond loss of control, including preventing increasingly capable AI systems from being misused by malicious or incautious actors. The solutions needed to monitor, evaluate, and if necessary enforce restrictions on frontier AI development will be critical to allow America to address risks from terrorism, proliferation of dangerous capabilities, and unauthorized use.

Building this comprehensive governance capacity now serves American interests regardless of one's assessment of extinction risk, as it provides policymakers with flexible options for responding to a wide range of AI development scenarios across varying timescales and risk profiles.

1.2 Other Risks

Beyond human extinction, our policy recommendations are responsive to two additional risks which threaten American national security: *human misuse* and *geopolitical destabilization*. Human misuse is when bad actors use AI systems to accomplish tasks they would not be able to otherwise, and includes scenarios such as terrorists developing novel pathogens. Geopolitical destabilization is the risk of tension, conflict, and war brought about by the impacts of AI on areas of [strategic significance](#) to the United States and its rivals, including warfare, cyberoperations, human persuasion, and intelligence gathering and analysis. The imagined impacts on these areas are so large that the expectation itself could lead to preemptive escalation. In particular, we should be prepared for those powers which expect to lose the race to ASI to [attempt](#) to [upset](#) the race before it can be completed. Any complete strategy needs to address this difficult dynamic.

Human misuse and geopolitical destabilization threaten widespread and severe damage to human well-being, potentially on a global scale. We expect these consequences earlier than ASI, and they are more salient to many than the threat of extinction. Fortunately, the policy required to address these threats overlaps with the policy required to address the threat of human extinction. Furthermore, our policy recommendations promote America's AI leadership and protect American innovation, as these

are critical advantages which must be safeguarded by the U.S. government and key to the creation of an effective off-switch. Foremost among these is the need to expand the capacity of the executive branch to reason about the complex issues involving AI development.

2. Recommendation: Expand State Capacity for AI Strategy

Create the National AI Strategy Office to expand U.S. government state capacity to keep pace with AI development, and to coordinate between agencies on these pillars:

- **Awareness:** assessing current and future AI capabilities
- **Readiness:** preparing for anticipated effects of AI systems
- **Emergency planning:** establishing protocols to respond to urgent AI threats

For simplicity, we'll discuss these state capacities as if they were to all be housed under the National AI Strategy Office. However, we are relatively agnostic about whether these powers and responsibilities should be under this new office, given to an existing agency, or some form of interagency coordinating body. The current U.S. AI Safety Institute (AIS) has developed expertise in evaluating AI models, but the scope of its responsibilities would need to be expanded to satisfy the critical capacities below.

The U.S. government must be able to keep pace with new AI technology. It is essential that within the government there is expertise to understand, monitor, and respond to new developments. As AI systems become increasingly capable and potentially surpass human-level intelligence within the next four years, the federal government must develop institutional knowledge and technical expertise to ensure America maintains its technological leadership. The government must enhance in-house capacity for understanding AI development.

Such measures have historical precedent. In the last 80 years, the United States has established sophisticated institutional frameworks to manage nuclear technology's immense potential and risks, including the Department of Energy's National Nuclear Security Administration, the Defense Threat Reduction Agency, and specialized units within STRATCOM. Similarly, the government now must establish specialized entities with deep technical understanding of AI systems.

Centrally, the government must develop the capacity to monitor, evaluate and, if necessary, intervene in the development of advanced AI systems as they become increasingly powerful and autonomous.

To effectively govern advanced AI development, the federal government must establish capacity across three essential pillars: **awareness** to accurately assess current and future AI capabilities; **readiness** to develop proactive policies in response to AI developments; and **emergency planning** to prepare coordinated responses for high-stakes scenarios.

2.1 Awareness

The first pillar, awareness, requires the government to develop the capacity for understanding the current state of AI and forecasting future developments. The proposed National AI Strategy Office should function as the central hub for gathering and analyzing critical information about AI capabilities and distributing it to relevant government agencies. To build effective awareness, this office should:

1. **Develop deep technical expertise** with frontier AI models, including their capabilities, limitations, and potential national security implications. This expertise would enable the government to evaluate and understand claims about AI capabilities. The office should establish strong working relationships with AI developers to facilitate information sharing about emerging capabilities and risks. The office may leverage expertise from the private sector to conduct evaluations.
2. **Establish robust evaluation frameworks** by expanding initiatives like the AISI to comprehensively assess the capabilities of advanced AI models. This should include assessment of capabilities relevant to national security, such as has been established with the [Testing Risks of AI for National Security \(TRAINS\) Taskforce](#). AI systems may also disrupt and automate human employment; capabilities here should be tracked so they can be appropriately managed.
3. **Monitor international AI developments** through coordination with intelligence agencies and diplomatic channels, ensuring the U.S. maintains accurate awareness of capabilities being developed by both allies and adversaries.
4. **Forecast future AI capabilities** before they materialize, enabling proactive rather than reactive policy development. This would include tracking key metrics like compute usage, model size, and performance improvements across public and private research.
5. **Establish transparency requirements for AI developers** to ensure visibility into frontier AI progress. These should include the reporting of training of frontier AI systems, disclosure of safety evaluation results, and comprehensive tracking of compute clusters and data centers involved in AI development. Tracking the location, capacity, and security of these physical computing resources is essential for understanding the AI development landscape. This reporting is very unlikely to be burdensome to AI developers; current reporting requirements can take about [one person-day to complete](#).
6. **Implement a whistleblower hotline and whistleblower protections**, modeled after the Nuclear Regulatory Commission's whistleblower hotline, allowing AI researchers to report concerns without fear of retaliation. This system would target concerns relevant to the risks we've described: the irresponsible development of AI systems with dangerous dual-use capabilities or technical safety issues relevant to the control of powerful AI systems.

2.2 Readiness

The second pillar, readiness, focuses on preparing for the effect of AI systems, rather than responding to crises after they occur. While awareness provides the foundation for understanding current and future AI capabilities, readiness involves developing concrete protocols to manage these capabilities.

1. **Conduct threat modeling** based on current and forecasted AI capabilities. This would include analyzing potential security vulnerabilities, weaponization pathways, economic disruption scenarios, and other large-scale risks from AI.
2. **Develop scenario planning frameworks** that anticipate potential challenges before they become emergencies. This includes working with various actors to identify interventions, governance mechanisms, and coordination protocols for managing emerging AI risks. The goal should be to identify inflection points where policy action is necessary, and implement responses before crises occur.
3. **Provide research and policy recommendations** to the government on critical issues including:
 - a. **The security and public safety implications of publicly releasing AI model weights.** If AI model capabilities continue to increase, it will [no longer be acceptable](#) to release their weights, due to concerns about misuse by malicious actors or empowering state adversaries. Current models likely fall below this threshold.
 - b. **Appropriate timing for implementing enhanced security measures based on capability forecasts.** Security measures may be required to prevent the theft of AI model weights and other important IP. The Office could provide policy recommendations about when the implementation of these measures needs to start, based on forecasts of AI model capabilities and forecasts of implementation times.
 - c. **Red-lines for AI development.** Are there AI capabilities which are too dangerous to develop? For example, advanced AI systems may enable the development of new WMDs, such as novel pathogens, and access to such systems should be restricted.
 - d. **How to monitor and address the safety and security of AI research within AI companies.** This research may become increasingly automated, and it is essential that humans still have control, oversight, and understanding.

2.3 Emergency Planning

The third pillar is emergency planning. Despite the efforts at awareness and readiness, unforeseen and urgent scenarios involving advanced AI systems may still arise. The government must have protocols in place to respond rapidly to novel threats and contain potential harms. To build effective emergency planning capacity, the National AI Strategy Office should:

1. **Develop comprehensive emergency response protocols** for AI-related crises including



malicious use, rogue autonomous system malfunctions, and the leakage/theft of AI model weights. These protocols should define clear lines of authority, communication channels, and response procedures that can be activated immediately when needed.

2. **Coordinate cross-agency responses** for AI-enabled threats, particularly in high-risk domains such as biological security, cybersecurity, and critical infrastructure protection.
3. **Establish shutdown protocols** for AI systems that present acute dangers. This includes developing technical mechanisms to safely turn off AI systems (e.g., shutting down a compute cluster), legal frameworks to authorize such interventions, and coordination procedures with private operators of AI infrastructure.

3. Recommendation: Maintain America's AI Leadership

Recommendations:

- Expand AI chip export controls to ensure U.S. leadership in AI.
- Improve security for AI model weights and algorithmic secrets to prevent theft by adversaries.
- Fund work into verification mechanisms to enable better governance of global AI activities.

3.1 Export Controls

A key component of ensuring America's lead in AI is strengthening existing export controls on AI chips, especially preventing their export to U.S. adversaries. Export controls contribute to an AI off-switch by centralizing compute to the US and its allies, likely making it easier to implement temporary controls on AI development. The Diffusion Framework and existing export control policy are positive steps in this direction, but more is needed. We have specific recommendations for improving the Diffusion Framework, which we will address in an upcoming comment to BIS; they include lowering company and country compute allocations and expanding the definition of AI chips (current definitions do little to limit U.S. adversaries' access to inference-viable AI chips). We discuss other recommendations here.

3.1.1 Expand export controls to other key manufacturing components and countries

Current export controls may have gaps that allow China to continue building advanced semiconductor manufacturing capacity. Components critical for EUV lithography are produced by two German companies, optical systems from Carl Zeiss and high-power lasers from TRUMPF. These German manufacturers continue substantial sales to China, potentially accelerating China's ability to develop domestic EUV capacity. The [Zeiss 2023/2024 Annual Report](#) states that they are seeing a growth in Semiconductor Manufacturing Technology due to "effects such as the strong demand for equipment for the manufacture of semiconductors in China".

It's unclear to us if these companies currently supply China with equipment for advanced AI chip production, but they both have the potential to, and are likely not prevented by existing European export controls. To ensure America's AI advantage, export controls should be expanded to cover equipment or components which are used for sub-10nm or EUV processes. The U.S. should work with Germany to implement export license requirements for this equipment (such as from TRUMPF or Zeiss), similar to existing restrictions on ASML's EUV machines from the Netherlands. This licensing process should include rigorous end-use verification and post-shipment monitoring to prevent diversion to restricted entities.

3.1.2 Chip registry

Chinese entities are [reportedly](#) smuggling thousands of AI chips in violation of U.S. export controls. To remedy this issue, we suggest the implementation of a registry for advanced AI chips in order to track chips from their fabrication (e.g., by TSMC) to their final data centers, and regular audits of data centers to ensure chips remain where they should be. [Previous work](#) has suggested various implementation details of such a registry, such as being run by BIS and involving random sampling to decrease costs. These registries should cover all advanced AI chips sold globally, including those residing in U.S. data centers. Ongoing audits or inspections may be crucial to ensure chips are not removed from their designated data center without notice, and reselling of chips would have to be tracked in the registry.

3.1.3 Security for AI model weights and algorithmic secrets

Beyond restricting access to large numbers of AI chips, effective export controls also require securing frontier AI model weights and key algorithmic secrets, as these are complementary to compute. Algorithmic progress internal to AI labs is moving at a staggering rate. AI systems require orders of magnitude less compute to train than they would have just a few years ago. For this reason, strategies to regulate hardware directly will be substantially weakened in the case that algorithmic secrets (including model architectures, training algorithms, optimizations, hyperparameters, and more) are not properly secured. Techniques to secure these details are under-researched, and we do not have more specific comments at this time. However, we expect some precedent for these measures exists in other fields, such as securing nuclear secrets.

Security for both model weights and algorithmic secrets could be [incentivized](#) through numerous approaches. For example, the U.S. government could provide tax incentives to encourage companies to invest in better security or could offer government procurement contracts predicated on meeting particular security standards.

The importance of securing model weights and algorithmic secrets is not obvious to all. Some argue

that external innovation such as DeepSeek’s V3 and R1 models shows the futility of effort here. We believe this is not the case as U.S. companies enjoy a compute advantage which they will use to produce the most capable models, and algorithmic progress is likely to be a major driver of AI capabilities progress in the future. We may elaborate on these arguments in upcoming work.

3.2 Governing AI Activities Globally with Verification Mechanisms

Frontier AI development (i.e., the largest AI training runs) should be kept in the United States and our close allies. Serving AI models to customers, on the other hand, should be a global endeavor supported by data centers around the world. However, such activities come with risks such as AI chips being smuggled to U.S. adversaries or AI models being misused for harmful activities. Fortunately, new mechanisms are being explored which could enable the U.S., in coordination with its allies, to govern AI activities globally in a light-touch manner.

We have [previously written in depth](#) about the mechanisms that could be used to verify compliance for international AI agreements. These mechanisms could serve many useful roles, including domestic and international governance of AI activities. For example, they could help ensure export-controlled chips stay where they are supposed to be, assist in evaluation of models for extreme risks, and ensure compliance with a bilateral US-China agreement. We highlight a few of these methods for which research and development must begin early. We encourage agencies such as the National Science Foundation and Defense Advanced Research Projects Agency to fund research on these AI verification mechanisms.

FlexHEG Mechanisms. Recently, researchers have conceptualized [Flexible Hardware Enabled Guarantee \(FlexHEG\) mechanisms](#), which could allow the governance of AI development in a precise and privacy preserving manner. These mechanisms would operate at the level of AI chips themselves (i.e., GPUs) or the server layer and would enable secure execution of governance functions. For example, one implementation involves a secondary computer chip connected to the main AI chip and placed in an enclosure to prevent tampering. This secondary chip enables various governance functions including, for example, verifying the truth of a developer’s model capability evaluations, tracking the total amount of operations (i.e., FLOPs) used to train an AI model, ensuring a company is serving the correct model to users rather than a fraudulent model, and much more. Importantly, FlexHEG mechanisms would be privacy-preserving (not leaking sensitive information to regulators). Currently, FlexHEGs are a conceptual framework with early prototypes, rather than a technical solution that could be implemented at scale. Further work can be funded to develop these mechanisms, including improving their security and scalability.

Privacy-preserving inference monitoring. When serving AI models to customers, AI developers often

run [secondary monitors](#) on the inputs and outputs of AI models in order to detect misuse or harmful behaviors. For example, these classifiers might aim to detect Child Sexual Abuse Material (CSAM) or AI models being used to [generate propaganda by U.S. adversaries](#). Similar monitors could be applied in an international context to allow light-touch monitoring of AI inference. Optimistically, technology development here could enable privacy-preserving monitoring that focuses only on the most extreme risks (e.g., development of biological weapons) without being a major concern for sovereignty and privacy. There are numerous political details to such monitoring that would need to be determined on a case-by-case basis, and funding early technical work could enable better options for privacy-preserving monitoring.

Inference-only AI chips. AI chips (e.g., GPUs, TPUs) are relatively general purpose, being capable of carrying out many kinds of AI workloads. In the future, it may be necessary to monitor or restrict what operations AI chips can do. Concretely, the Diffusion Framework aims to keep frontier AI development in the U.S. and close allies, while allowing much of the world to carry out AI inference and smaller-scale AI training. Ideally, specific AI chips could help implement such a goal via being *capable of inference but not training*. Such chips could then be sold at scale without concern about them being reappropriated for training. Currently, many chips that are marketed for inference (e.g., Amazon's [Inferentia](#)) can still be used for training very [effectively](#). Therefore, more work is needed to both define and build inference-only chips. Inference-only chips could be a win-win, empowering the global community to benefit from AI advancements without posing risks to U.S. leadership or national security. Unfortunately, inference itself might eventually pose these risks, rather than just training, so this approach may not work in perpetuity.

4. Recommendation: Coordinate With China to Ensure Stability

Recommendations:

- Develop intelligence capabilities for AI
- Coordinate with China
- Anticipate diplomatic needs should we end up in a confrontation
- Reinforce communication channels and build trust incrementally

4.1 Develop intelligence capabilities for AI

To ensure its own ability to respond appropriately to developments, the U.S. government should invest in the zero-trust verification mechanisms required to maintain intelligence over foreign efforts to develop frontier AI. This includes many complex activities such as satellite imagery, signals intelligence, and cyber intelligence. The applicability of zero-trust mechanisms should be clear, given the

deteriorating relationship between the U.S. and China, in particular. They can also be a key component of other efforts, such as appraising the effectiveness of export controls.

4.2 Coordinate with China

Furthermore, the U.S. should begin the conversation with China to coordinate on which zero-trust verification mechanisms are redlined and which are not. As China will likewise rely on its own capabilities to assess the state of American AI developers, it is crucial that we develop a mutual understanding of which mechanisms will be acceptable and which will not.

Consider the [ABM treaty of 1972](#), which, in Article XII, codified the use of satellite surveillance between the U.S. and the U.S.S.R. It further prohibited deliberate concealment measures that would impede verification. It may be important to identify and coordinate which surveillance techniques applicable to AI development will not be impeded as we approach strategically significant AI.

It is important that China not feel great uncertainty about the state of American AI progress, as America holds a position of leadership in frontier AI development. [China will likely be attuned to the key AI risks we highlight, including human extinction.](#) As an example, Wen Gao is director of a major Chinese AI lab and dean at Peking University. He has influence in technical and political circles in China, serves on high-level advisory bodies for the government, and led the Politburo in a collective study session on AI. In a 2021 paper he raised concerns about how AGI has the potential to, through recursive self-improvement, lead to an intelligence explosion and catastrophe.

In addition to these risks, China has reason to worry about the strategic consequences of advanced AI. Anthropic CEO Dario Amodei [has been explicit](#) about his intention to have the fruits of AI development help “U.S. and its allies ... take a commanding and long-lasting lead on the global stage.” To be clear, his implication is that this lead is not just in AI but across all endeavors. As attractive as that outcome is for America, it will be gravely concerning for China.

We should expect China to use its intelligence gathering capabilities to monitor the progress of leading American AI labs and to use that intelligence to guide its actions. If the intelligence is poor, the actions will be more severe and unpredictable. Thus we come to the perhaps counterintuitive result: we advocate **against** policies which undermine China’s ability to accurately monitor American AI progress. A relevant distinction can be made between concealing these activities from intelligence gathering (undesirable) vs. securing the fruits from theft or sabotage (desirable).

Another historical parallel is the 1992 Treaty on Open Skies, which the U.S., Russia, and 32 other countries entered into to facilitate mutual intelligence gathering on military forces and activities. By allowing countries to openly surveil each other, this treaty aimed to prevent misunderstandings and

limit tensions.

4.3 Anticipate diplomatic needs

If and when China takes action on AI risk, what form will it take, and how can the U.S. best prepare? We should prepare for China to approach the U.S. to seek a diplomatic resolution to their concerns.

If China is most concerned with human extinction or misuse risks, the U.S. government may need to provide assurances. Doing this requires that leading AI companies have invested in the appropriate research or security techniques, so we recommend the U.S. government support those activities, through subsidy or incentive. Assurances will also require the U.S. government's own tracking of the situation through expanded state capacity.

On the other hand, what if China is most concerned about losing the race to strategically decisive AI? The most desirable solution is the ability to make credible commitments for benefit sharing. That is, perhaps the U.S. can allay these fears through promises and use of [a suite of international relations credibility tools](#). That said, the actual durability of credible commitments in the face of a looming decisive strategic advantage is an unresolved question.

Ultimately, it may be necessary for both parties to halt AI development for at least long enough for risks and fears to be addressed. We expect mutual independent validation mechanisms to play a substantial role in an AI development halt, extending the significance of coordination on this topic.

4.4 Reinforce Communication Channels and Build Trust Incrementally

In an emergency, it could be useful to have a reliable and direct channel of communication. While a US-China presidential-level hotline exists, China has an inconsistent history of answering it. [See this report for a more thorough discussion](#). We recommend effort to resolve the cause of this, for its broad potential to be useful in domains including but not limited to AI. Additional channels of communication between the U.S. and China should be developed or reinforced. These include cabinet or staff-level links, military-to-military crisis channels, and Track II (or citizen) dialogues.

Trust can be built incrementally, and small commitments which are satisfied can lead to larger commitments. We should seek areas where cooperation is possible today that might enable cooperation on more critical issues in the future. This could include, for example, coordinated efforts to block the release of open weight models beyond a level of dangerous capabilities by companies in either country as we discuss in the next section.

5. Recommendation: Restrict Proliferation of Dangerous AI Models



Recommendations:

- Pilot “differential access” programs allowing early access to AI models for security and preparedness.
- Conduct ongoing assessment of risks and benefits of open source AI to enable future recommendations.
- For now, prohibit the open release of AI models that would score “High” on the OpenAI Preparedness Framework at any point in their development cycle.

Open sourcing in AI development commonly refers to the release of model weights, but could also include research insights and, less commonly, the actual code (and data) required to recreate the model. Open sourcing has benefits such as enabling better third-party testing, supporting the broader research ecosystem, and making it easier for companies to build on AI products.

However, open sourcing also poses risks. One specific risk from open sourcing AI design information or AI model weights is enabling rogue actors to accomplish extremely dangerous tasks, such as creating novel biological weapons. This stems from the field’s lack of robust techniques for preventing the misuse of the capabilities such models offer. While AIs are not currently able to enable such threats, [model evaluations](#) and forecasting indicate that AI systems may be sufficiently capable in the next four years, exacerbating threats to U.S. national security.

A second specific risk is the proliferation of AI models that can accelerate AI research among less-resourced (and therefore less-detectable) actors. The proliferation of such systems, if they are developed in the future, would make it effectively impossible to enforce basic AI regulations or implement a global off-switch, even if there were imminent risk of catastrophic harm. Besides model weights, it may be desirable to eventually limit the release of other key AI insights, such as model architectures, training algorithms, hyperparameters, datasets and RL environments.

It is likely feasible to prevent (or at least significantly limit) the proactive release of critical AI intellectual property, including model weights, research insights, and code. There is precedent for restricting the release of harmful information in other domains, such as nuclear security and weapons development. We encourage exploring the application of this precedent to protect AI secrets. These AI secrets could be critical to U.S. national security due to risks from rogue actors and U.S. adversaries.

We offer three specific suggestions to reduce risks from open source AI systems:

Pilot “differential access” programs starting in 2025. Such programs could involve [structured access](#) and [staged release](#), for instance releasing a private API for specific defensive actors, then a public API, and finally open release of model weights, depending on the risks. These programs should allow early



access to frontier AI models for organizations working on critical security and preparedness. For example, they could include cybersecurity teams at major tech companies (such as [Google's Project Naptime](#)), nuclear security experts in DoE, and biological security experts [across](#) government. As AI systems get more powerful, it could be crucial that defenders have more time with such systems in order to detect and remedy vulnerabilities, before such AI systems are widely available to attackers. Because advanced AI capabilities could be developed soon, it may be crucial to start an early access program in 2025 so the relevant actors have time to adopt such a system into their work and resolve any difficulties. Applying differential access to AI is the topic of forthcoming work from Shaun Ee and collaborators, and it has been [discussed](#) previously.

One potential way to facilitate differential access is through the [National AI Research Resource](#) (NAIRR). The NAIRR could include secure compute infrastructure (i.e., one or more high security AI data centers) that allows for third-party use and testing without posing major security risks. For instance, AI developers could give their models to the NAIRR for hosting, and NAIRR could then provide access to these models for vetted researchers, without incurring risks from publicly releasing AI model weights. However, NAIRR needs more funding in order to be effective.

The National AI Strategy Office (described above) should conduct ongoing evaluation of the risks and benefits of open-source in the AI space. This Office should make regular recommendations and provide expert guidance to other parts of government about this topic. Its mandate should include the release of model weights, algorithmic secrets (e.g., model architectures, training algorithms, hyperparameters), and data (e.g., RL environments, high quality reasoning datasets, datasets pertaining to dual-use capabilities such as biosecurity and cybersecurity). The Office should be primarily focused on technical details and should work closely with chemical, biological, radiological, and nuclear (CBRN) and cyber experts in order to understand AI model capabilities. The Office would need substantial technical expertise.

Before full guidance has been developed, **implement interim rules to avoid catastrophic outcomes from open weight model release**. While specifics are being developed, default to prohibiting the open release of models that would score High on any category in [OpenAI's preparedness framework](#) (Cybersecurity, CBRN threats, Persuasion, Model autonomy). This capability level corresponds approximately to a human expert in the relevant domain (i.e., 90th percentile among relevant professionals). Concretely, such a model may be capable of helping an expert to engineer a novel CBRN threat vector, enabling someone with only basic training to create a serious chemical or biological weapon, or generating persuasive content comparable to a national-level influence agent. As of writing, the [Deep Research](#) system developed by OpenAI is likely the closest publicly known system to this threshold, scoring Medium in all four categories.

We think it is likely that the first AI system to pass a High threshold will be developed in 2025 and that there would be an open release of such a model in 2026, barring intervention. This evaluation should be carried out as a form of pre-deployment testing by an independent body, such as the National AI Strategy Office and/or the US AISI, with relevant expertise. This judgment should be based on the highest scoring version of a model and should not take into account “mitigations” by the developer. Principally, this is because all known mitigation strategies can be [easily removed](#) for open-weight models. If there is substantial progress in research on mitigations, it may become possible to robustly mitigate risk from otherwise-dangerous models whose weights are released, but current methods are not sufficient. Because of the rapidly changing landscape, the authority to prohibit open release of such models should rest with the executive branch, informed by the work of these independent bodies.

Conclusion

The risks posed by advanced AI systems—specifically human extinction, misuse, and geopolitical destabilization—demand urgent, coordinated action. The United States must lead a globally coordinated effort to build and maintain an off-switch for AI development while simultaneously securing its technological lead. By expanding state capacity for AI strategy, maintaining America’s AI leadership, coordinating with China to ensure stability, and restricting the proliferation of dangerous AI models, the U.S. can be better prepared for the complex challenges ahead. These recommendations serve both immediate policy objectives and preserve critical options for the future. As AI systems approach and potentially surpass human-level intelligence within the next few years, the actions we take today will determine whether this transformative technology serves humanity’s interests or threatens our existence. We urge the administration to act decisively while the window of opportunity remains open, implementing and further building upon these recommendations to ensure AI development proceeds in a manner that is secure and beneficial to all Americans.

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